

4.4 Setup & Installation Guide

4.4.1 Overview

CampusCare consists of two main parts:

Component	Technology
Backend API	Node.js + Express
Mobile Application	React Native + Expo

The system uses:

- Supabase PostgreSQL database
- Supabase Storage for cloud image uploads
- JWT authentication for security

4.4.2 Prerequisites

Before running the project, install the following:

Software	Purpose
Node.js (v18+)	Run backend and frontend
npm	Install dependencies
VS Code	Code editor
Expo Go	Run app on mobile
Git	Version control
Supabase Account	Database and storage

Check installed versions:

```
node -v
npm -v
```

4.4.3 Project Structure

```
campuscare-project/
├── backend/
└── mobile/
```

4.4.4 Backend Setup

Step 1: Open Backend Folder

```
cd backend
```

Step 2: Install Dependencies

```
npm install
```

Step 3: Create .env File

Create a file named:

```
.env
```

Add:

```
PORT=3000
SUPABASE_URL=your_supabase_url
SUPABASE_ANON_KEY=your_supabase_anon_key
JWT_SECRET=campuscare_secret
```

Step 4: Run Backend

```
npm run dev
```

Expected output:

```
Server running on port 3000
```

4.4.5 Supabase Setup

Step 1: Create Supabase Project

Create a project named:

```
CampusCare
```

Step 2: Create Tables

Create the following tables:

- users
- issues
- comments

- `issue_photos`

Step 3: Create Storage Bucket

Go to:

Storage → New Bucket

Create bucket:

`issue-photos`

Set bucket to public.

4.4.6 Frontend Mobile Setup

Step 1: Open Mobile Folder

```
cd mobile
```

Step 2: Install Dependencies

```
npm install
```

Step 3: Configure API URL

Open:

```
mobile/src/api/api.js
```

Set backend URL:

```
baseUrl: "http://YOUR_IP:3000/api"
```

Example:

```
baseUrl: "http://192.168.1.206:3000/api"
```

Step 4: Run Expo

```
npx expo start -c
```

Scan the QR code using Expo Go.

4.4.7 Find Laptop IP Address

Run:

```
ipconfig
```

Copy:

```
IPv4 Address
```

Example:

```
192.168.1.206
```

The phone and laptop must be connected to the same Wi-Fi network.

4.4.8 Test Accounts

Role	Email	Password
Community Member	testuser@gmail.com	123456
Manager	manager1@gmail.com	123456
Worker	worker1@giu.edu.eg	123456

4.4.9 Full Workflow Testing

Community Member

- Login
- Submit issue
- Upload issue photo
- View submitted issues

Facility Manager

- View all issues
- Assign worker
- Update issue status

Worker

- View assigned issues
- Add completion comment
- Upload completion photo
- Mark issue resolved

4.4.10 Common Issues

Problem	Solution
App cannot connect	Update IP address in <code>api.js</code>
401 Unauthorized	Login again and use valid JWT token
403 Forbidden	User role not allowed
Image upload fails	Check Supabase bucket and credentials
No workers found	Ensure users with role <code>worker</code> exist

4.4.11 Run Commands Summary

Backend

```
cd backend
npm run dev
```

Frontend

```
cd mobile
npx expo start -c
```

4.4.12 GitHub Repository

Project repository:

[CampusCare GitHub Repository](#)

Clone repository:

```
git clone https://github.com/salwakhaledmetwally0726-glitch/campuscare-project.git
```